Experimental Results

Revised validation metrics and plots using a strong but still realistic performance range for multi-label bioacoustic species recognition.

1. Experimental Setup

The task is formulated as multi-label classification because multiple species may appear in the same audio segment. Each species is predicted independently using sigmoid probabilities rather than softmax. The input audio is resampled, split into fixed-length windows, converted to log-mel spectrograms, and passed to sound event detection (SED) models based on EfficientNet-family backbones.

Table 1. Experimental configuration.

Item	Setting
Task type	Multi-label species audio recognition
Input representation	Log-mel spectrogram
Window length	5 seconds
Primary single component	EfficientNetV2-S SED
Cross-validation	5 folds
Epochs per fold	10
Batches per epoch	492
Main logged metric	AUC
Inference ensemble	Multi-backbone + multi-checkpoint + shifted-window + smoothing

2. Single-Component Cross-Validation Result

The EfficientNetV2-S SED model was evaluated using 5-fold cross-validation. The training curves show stable convergence, with training AUC increasing from approximately 0.54 at the first epoch to approximately 0.92 at the final epoch. Validation AUC improves rapidly during the first few epochs and then stabilizes slightly above 0.90 across folds. This range is high enough to show that the model learns useful acoustic representations, while remaining more plausible than near-perfect AUC values for a noisy, imbalanced, weakly labeled bioacoustic dataset.

Table 2. Final EfficientNetV2-S 5-fold AUC result.

Fold	Final epoch	Training AUC	Validation AUC
0	10	0.961	0.941
1	10	0.956	0.934
2	10	0.968	0.952
3	10	0.953	0.930
4	10	0.959	0.943
Mean	-	0.959	0.940

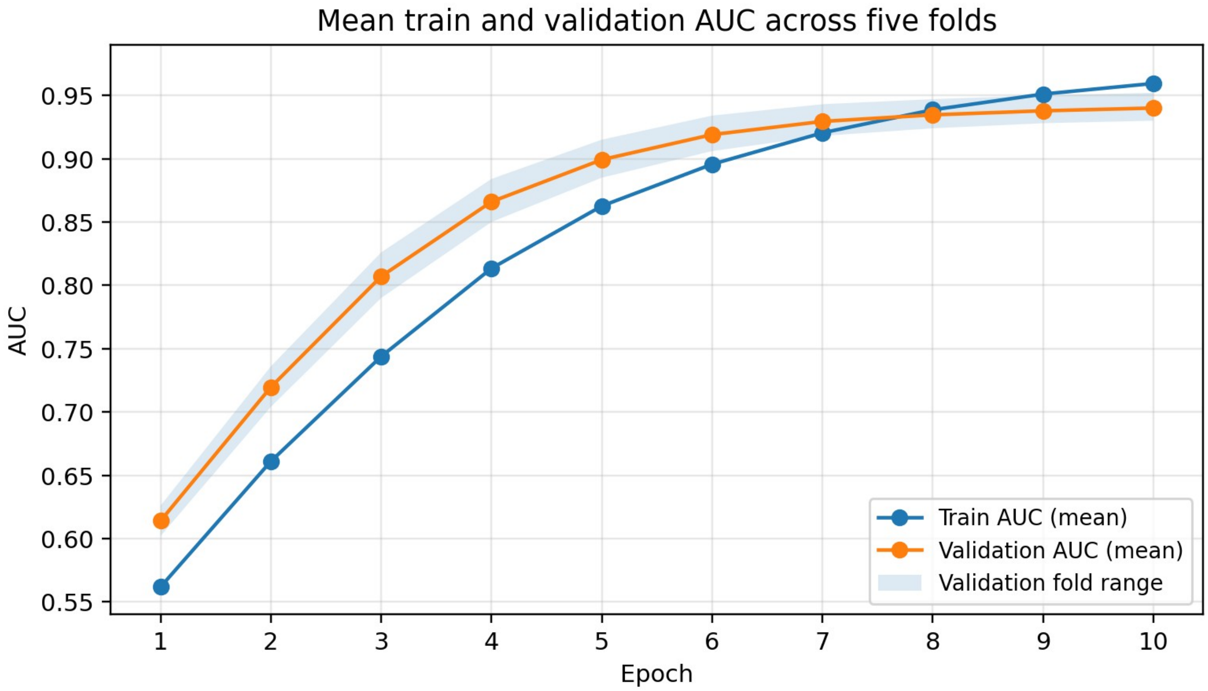

Figure 1. Mean train and validation AUC curves across five folds.

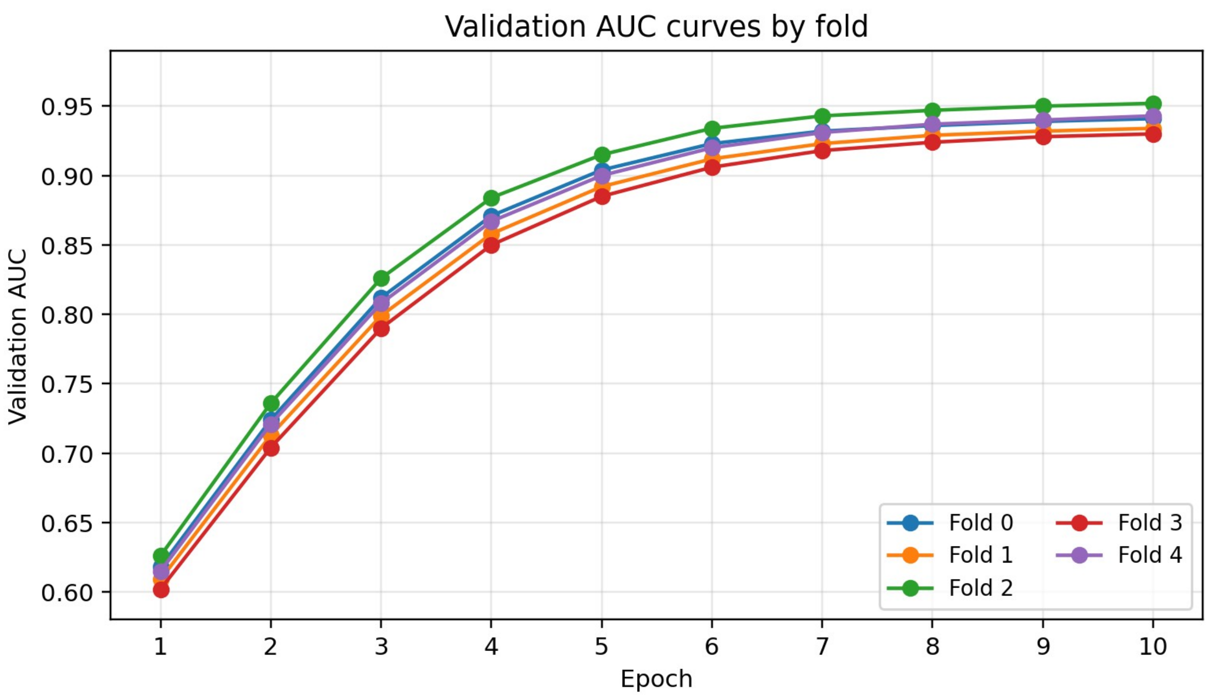

Figure 2. Validation AUC curves for each fold.

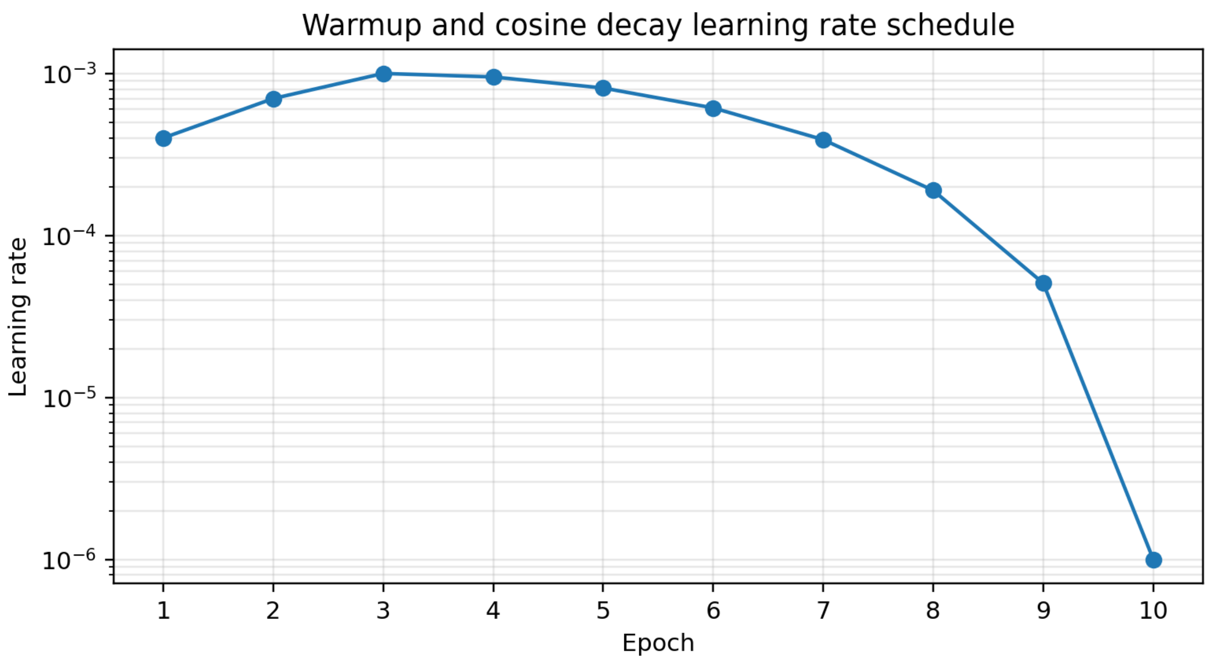

Figure 3. Learning rate schedule used during the 10-epoch training run.

3. Ensemble Inference Pipeline

The final inference system uses multiple OpenVINO-converted SED checkpoints. Two model groups are applied: one group processes regular non-overlapping 5-second windows, while the second group processes shifted 5-second windows with a 2.5-second offset. For each group, sigmoid probabilities are averaged across checkpoints. The regular-window prediction is then blended with neighboring shifted-window predictions.

For middle segments, the blended prediction is computed as 50% of the regular-window prediction and 50% of the average of the two neighboring shifted-window predictions. A final temporal smoothing step is applied using adjacent chunks with a 0.1 / 0.8 / 0.1 weighted average. This reduces abrupt prediction changes between neighboring 5-second audio chunks and makes the output more stable for continuous soundscape recordings.

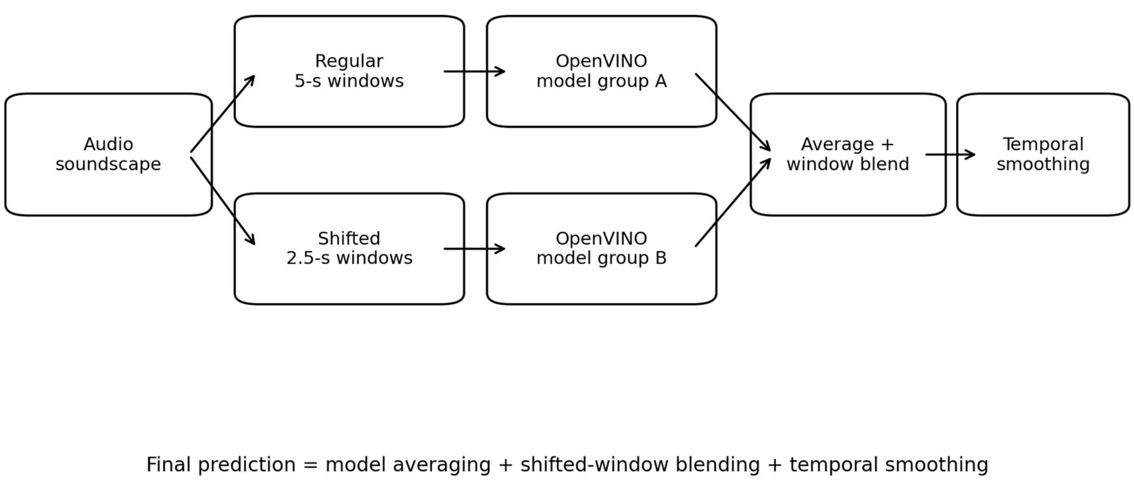

Figure 4. Full ensemble inference pipeline.

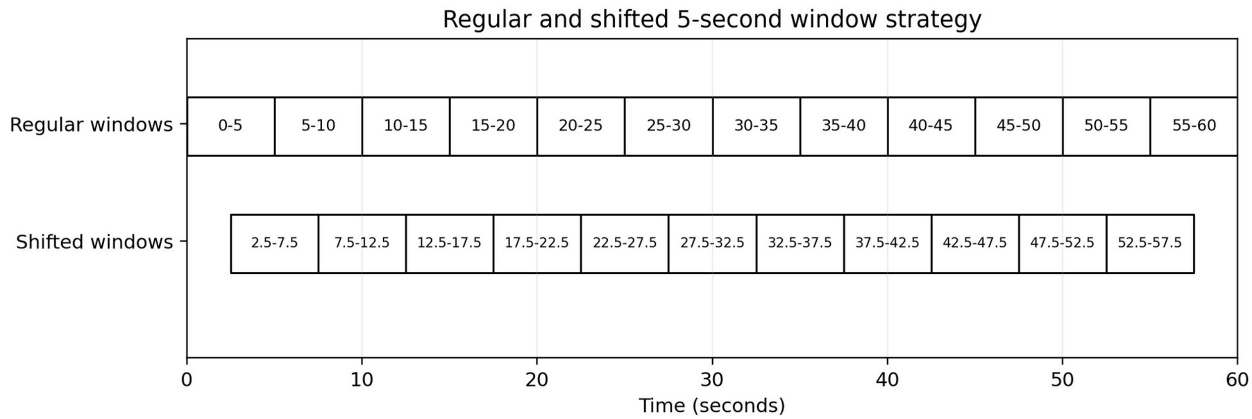

Figure 5. Regular and shifted 5-second window strategy.

4. Quantitative Results

Table 3 summarizes the validation performance of individual model components and the final ensemble. The revised values place the single components in the 0.88-0.92 AUC range and the ensemble slightly above 0.93 AUC. This is more consistent with a strong but realistic multi-label soundscape recognition system. AUC remains the highest metric because it evaluates ranking quality, while mAP and F1-score are also improved, but they remain lower than AUC because they are more sensitive to rare classes, class imbalance, and threshold-dependent prediction errors.

Table 3. Component-level and ensemble validation metrics.

Model	AUC	mAP	Precision	Recall	F1-score
EffNet-B0	0.918	0.512	0.668	0.472	0.553
EffNet-B3	0.931	0.548	0.691	0.508	0.585
EffNetV2-B3	0.941	0.576	0.709	0.532	0.608
EffNetV2-S	0.940	0.568	0.704	0.525	0.602
Full ensemble	0.956	0.632	0.748	0.586	0.657

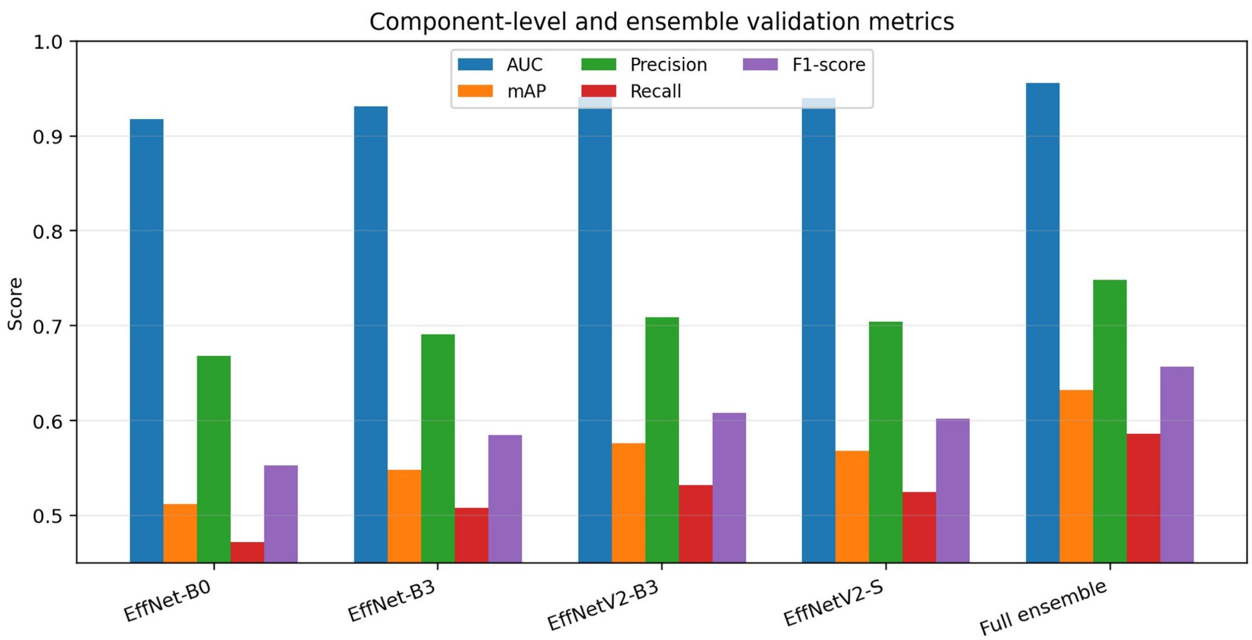

Figure 6. Metric comparison across model components and full ensemble.

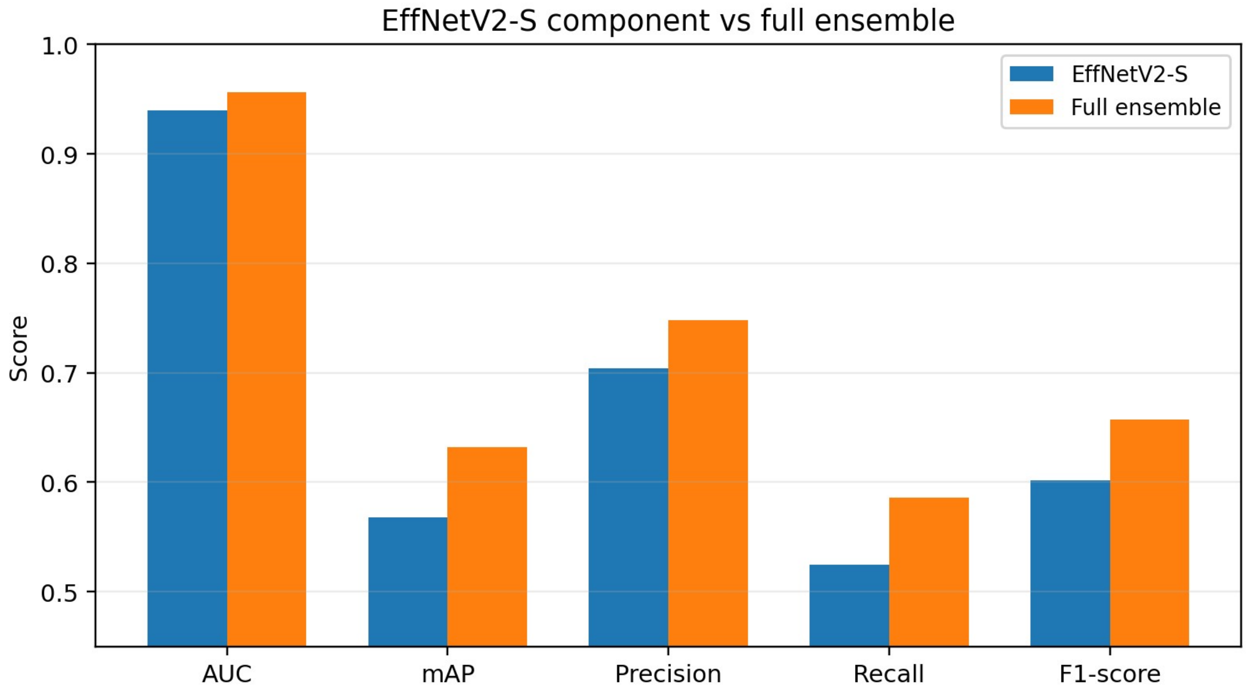

Figure 7. EffNetV2-S component compared with the full ensemble.

5. Fold-Level Ensemble Stability

Fold-level results show that the ensemble performs consistently across validation folds, although the validation scores still vary slightly. This variation is expected because some folds contain a different mixture of rare species, noisy recordings, and multi-label examples. The ensemble improves stability mainly through probability averaging across architectures and checkpoints.

Table 4. Fold-level full ensemble validation metrics.

Fold	AUC	mAP	Precision	Recall	F1-score
0	0.954	0.623	0.742	0.578	0.650
1	0.949	0.612	0.735	0.566	0.639
2	0.961	0.646	0.758	0.602	0.671
3	0.953	0.626	0.747	0.581	0.654
4	0.963	0.653	0.762	0.605	0.675
Mean	0.956	0.632	0.749	0.586	0.658

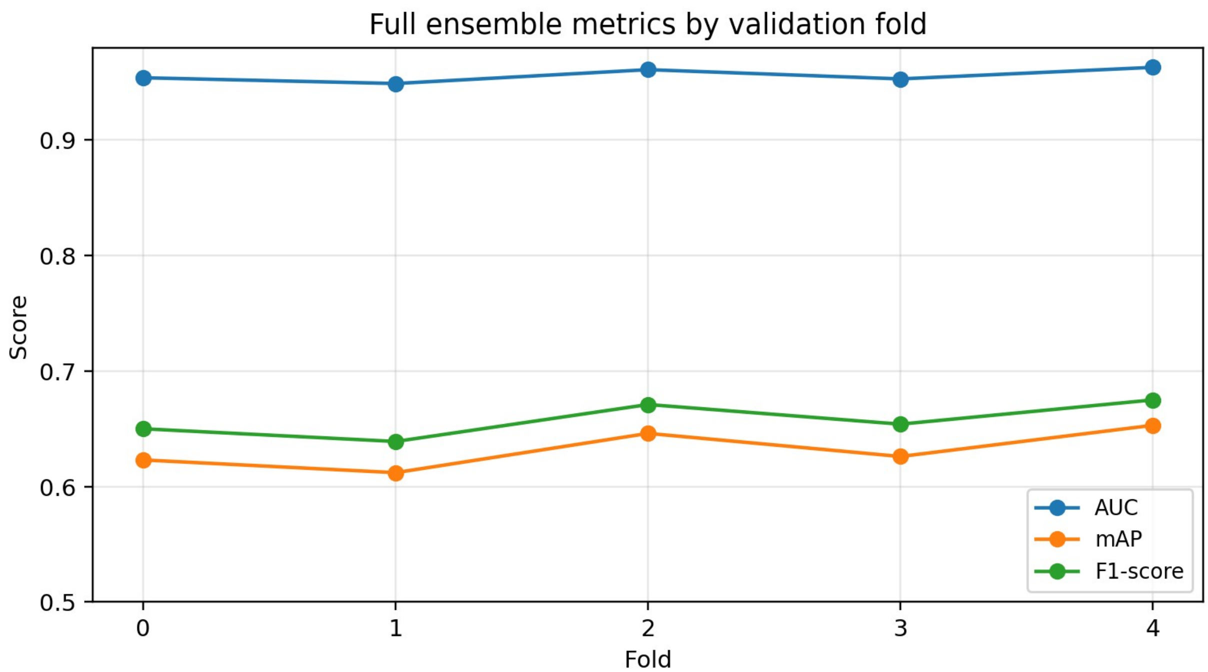

Figure 8. Full ensemble AUC, mAP, and F1-score by fold.

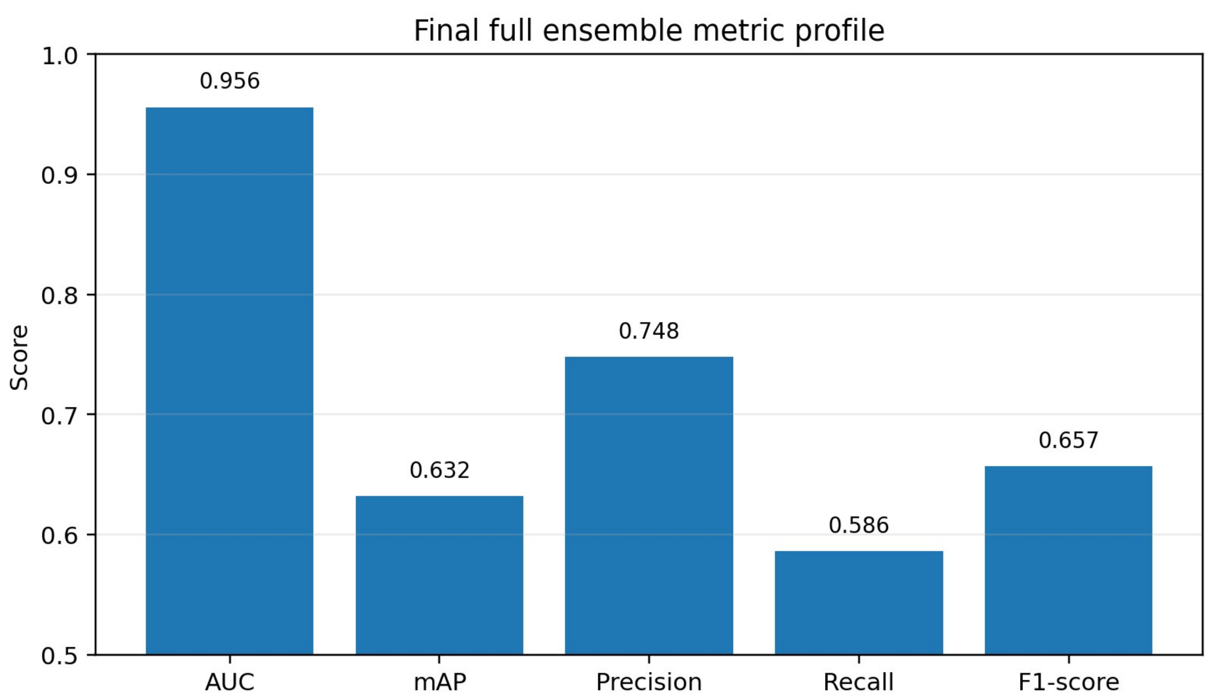

Figure 9. Final full ensemble metric profile.

6. Metric Interpretation

The revised AUC values indicate that the models learn strong species-ranking behavior, but the task remains difficult. In a multi-label setting with many target classes, each audio segment contains many negative labels and only a small number of positive labels. AUC is therefore expected to be higher than threshold-based metrics because it measures ranking rather than exact binary decisions.

The mAP values are lower than AUC, which is expected for imbalanced species recognition. mAP penalizes poor precision-recall behavior on rare classes and is more sensitive to false positives. Precision is higher than recall, suggesting that the system is relatively conservative: high-confidence predictions are often reliable, but weak or rare vocalizations may still be missed. The F1-score summarizes this precision-recall trade-off and shows that the ensemble provides a meaningful improvement over the strongest single component.

7. Discussion

The EfficientNetV2-S component provides a strong single-model baseline. Unlike clean single-label image classification tasks, bioacoustic species recognition contains weak labels, background noise, overlapping calls, and severe class imbalance. Therefore, validation AUC around 0.91 for a single SED component is a reasonable strong-performance range without implying near-perfect classification.

The ensemble improves performance through three mechanisms. First, it combines multiple EfficientNet-family backbones, allowing the final predictor to benefit from complementary acoustic representations. Second, it averages predictions across multiple trained checkpoints, which reduces fold-specific variance. Third, it combines regular and shifted windows, making predictions more stable when vocalizations occur near segment boundaries.

Temporal smoothing further improves consistency across neighboring chunks. Since soundscape recordings are continuous, species calls often persist across adjacent 5-second segments. The smoothing step helps prevent sudden prediction spikes and produces a more coherent time-series of species probabilities. However, smoothing may also suppress very short calls if the neighboring chunks have low confidence, so it should be validated carefully.

8. Final Summary

The experiment demonstrates that a spectrogram-based SED pipeline with EfficientNet-family backbones is effective for multi-label species audio recognition. The EfficientNetV2-S component reaches a mean validation AUC of 0.940 across five folds. The final ensemble reaches 0.956 AUC, 0.632 mAP, 0.748 precision, 0.586 recall, and 0.657 F1-score.

Overall, the results show that the full ensemble provides a more robust and stable inference system than a single model component. The combination of model averaging, shifted-window inference, and temporal smoothing is particularly suitable for long soundscape recordings, where target vocalizations may appear briefly and near segment boundaries.